```
.SUBCKT RUYy0_400k_120k_300MVA  1U  1V  1W  1N  2U  2V  2W  2N
+  PARAMS:
+  SN=300e6
+  U1Nenn=400e3
+  U20=120e3
+  fn={fn1}
+  N1=1000
+  uR=0.002
+  uX=0.2
+  i0=0.0006
+  p0=0.0004
+   a=0.75 ; FS
+   m=9 ; FS
+   psi_0=0 ; Remanence-parameter 0,1,-1
+   psir=1
+   psis=-0.5
+   psit=-0.5
+   tpsi0=1000s  ; Remanence-timepoint
+   tpsid=0.1ms  ; On-time of remanent flux
+   ron=0.00001  ; On - Parameter of switcher for remanent flux
+   bf=0        ; bf=1 -> middle leg with shorter magn. path correction factor
+   rkessel=6e5  ; Tank resistance
+   l_=0
+   middle=0
+   stproz=1.0
+   ctrl={1+(l_-middle)*stproz/100} ; simple Tap changer
Xu   1U   1N   2U   2N   magn1   magn2   Trafoschenkel2_Yy
+ PARAMS:
+ SN={SN/3} U1N={U1Nenn/sqrt(3)} U20={U20/sqrt(3)} fN={fN} N1={N1} uR={uR}
+ uX={uX} i0={i0*IF(bf==0,1,1.2)} p0={p0*IF(bf==0,1,1.2)} a={a} m={m}
+ psi_0={psir*psi_0} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl={ctrl}
Xv   1V1N   2V   2N   magn1   magn2   Trafoschenkel2_Yy
+ PARAMS:
+ SN={SN/3} U1N={U1Nenn/sqrt(3)} U20={U20/sqrt(3)} fN={fN} N1={N1} uR={uR}
+ uX={uX} i0={i0*IF(bf==0,1,0.6)} p0={p0*IF(bf==0,1,1.2)} a={a} m={m}
+ psi_0={psir*psi_0} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl={ctrl}
Xw   1W   1N   2W   2N   magn1   magn2   Trafoschenkel2_Yy
+ PARAMS:
+ SN={SN/3} U1N={U1Nenn/sqrt(3)} U20={U20/sqrt(3)} fN={fN} N1={N1} uR={uR}
+ uX={uX} i0={i0*IF(bf==0,1,1.2)} p0={p0*IF(bf==0,1,1.2)} a={a} m={m}
+ psi_0={psir*psi_0} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl={ctrl}
Rm   magn1   0   1MEG
Rkessel magn1 magn2 {rkessel}
RGND1U 1U 0 1e12
RGND1V 1V 0 1e12
RGND1W 1W 0 1e12
RGND2U 2U 0 1e12
RGND2V 2V 0 1e12
RGND2W 2W 0 1e12
RGND1N 1N 0 1e12
RGND2N 2N 0 1e12
.ENDS

.SUBCKT   Trafoschenkel2_Yy  1_1 1_2  2_1 2_2  magnP1 magnP2
+  PARAMS:
+  SN=10kVA
+  U1N=400V
```

```
+ U20=400V
+ fn={fn1}
+ N1=1000
+ uR=0.04
+ uX=0.08
+ i0=0.05
+ p0=0.005
+ a={0.3} ; FS
+ m={7} ; FS
+ psi_0=0 ; Remanenzparameter zwischen 0,1,-1
+ tpsi0=40ms
+ tpsid=0.1ms
+ ron=0.00001
+ ctrl=1
.PARAM
+ I1N={SN/U1N}
+ Z1N={U1N/I1N}
+ omegaN={(2*pi)*fN}
+ R1={Z1N*uR/2}
+ L1s={Z1N*uX/2/omegaN}
+ RFe={U1N*U1N/(p0*SN)}
+ N2={N1*U20/U1N}
+ R2={R1*N2/N1*N2/N1}
+ Xh={(U1N*U1N)/(SN*sqrt(i0*i0-p0*p0))} ;FS
+ iu={U1N/Xh} ;FS
+ ImyN={sqrt(2)*iu} ;FS
+ phiN={sqrt(2)*U1N/omegaN/N1}
+ a={a} ; FS
+ m={m} ; FS
* Windingscurrents
Gi1 1_1 1_2 VALUE={v(a,magnP1)/(N1*ctrl)}
Rpi1 1_1 1_2 1000MEG
Gi2 2_1 2_2 VALUE={v(b,a)/N2}
Rpi2 2_1 2_2 1000MEG
* Windings
X_W1 magnP1 a 1_1 1_2 winding_Yy
+PARAMS: R={R1} N={N1} Rphis={N1*N1/L1s}
+psi_0={psi_0*phiN} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl={ctrl}
X_W2 a b 2_1 2_2 winding_Yy
+PARAMS: R={R2} N={N2} Rphis={N1*N1/L1s}
+psi_0={psi_0*phiN} tpsi0={tpsi0} tpsid={tpsid} ron={ron} ctrl=1
* Iron leg
X_SCH b magnP2 Leg_Yy PARAMS:
+ N={N1} RFe={RFe} ImyN={ImyN} ctrl={ctrl}
+ phiN={phiN} a={a} m={m}
.ENDS

.SUBCKT winding_Yy ma mb ea eb
+PARAMS: R=1 N=1 Rphis=1 psi_0=0 tpsi0=1000s tpsid=0.1ms ron=0.00001 ctrl=1
Gphi ma mb VALUE={v(phi)}
Rphis ma mb {Rphis*ctrl*ctrl}
*********************************************************************************
Gp 0 phi VALUE={(v(ea,eb)-R*ctrl*v(mb,ma)/(N*ctrl))/(N*ctrl)}
Cp phi 0 1 IC=0
Rpphi 01000MEG
Vctrl ctrl 0 pwl 0s 0v {tpsi0+1p} 0v {tpsi0+tpsid+2p} 1V {tpsi0+2*tpsid+3p} 1v {tpsi0+3*tpsid+4p} 0v
S1 phi phix ctrl 0 MS1
```

```
.model MS1 SW(Ron={Ron} Roff=1e9 Vt=0.9 Vh=0 Lser=0 Vser=0)
E1 phix 0 value={psi_0}
.ENDS ;

* magnetic leg
.SUBCKT Leg_Yy  m1  m2
+PARAMS:   N=1  RFe=1  ImyN=1  a=1  phiN=1 m=9 ctrl=1
LRFem1    mh   {N*ctrl*N*ctrl/RFe} Rser={0.001m*N*N/RFe}  ; Rser=parasitic Value
vphi  mh    mh1  0
rphi    mh  0 1e12
Eimy mh1  m2
+VALUE={N*imyN*(a*i(vphi)/phiN+(1-a)*PWRS(i(vphi)/phiN,m))}
.ENDS;
```